

PROGRAMMING FUNDAMENTALS

Spring 2024

23rd May

Ex-1

- Use **while** loop structure to accept integer values in a 2-D array in the following format:

```
i = 0 and j = 0 :8
i = 0 and j = 1 :3
i = 0 and j = 2 :4
i = 1 and j = 0 :1
i = 1 and j = 1 :5
i = 1 and j = 2 :9
i = 2 and j = 0 :6
i = 2 and j = 1 :7
i = 2 and j = 2 :2
```

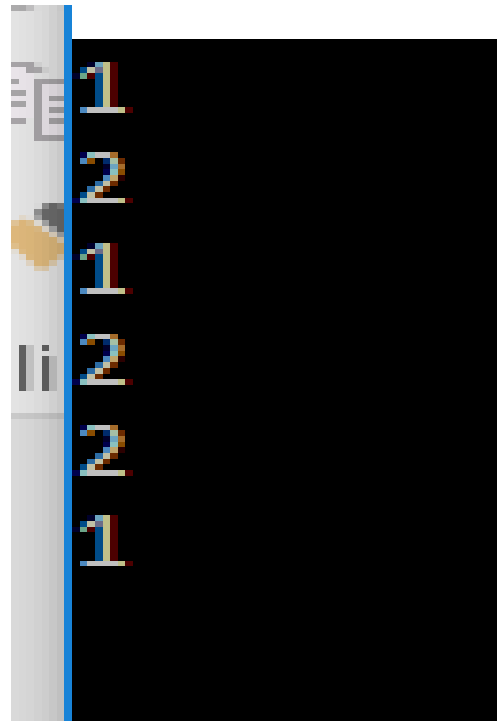
Ex-2

- Use **For** loop structure to display the array created in the previous question as below:

```
The Matrix is displayed below
8      3      4
1      5      9
6      7      2
```

Ex-3

- Write a **For** loop that generates five random integers between 1 and 2
- Sample output is shown below:

A screenshot of a code editor with a dark background. On the left, there is a vertical toolbar with icons for a list, a cursor, and a line of code. The main area displays a list of six integers: 1, 2, 1, 2, 2, 1. The numbers are arranged in a single column, with each number on a new line. The first number is 1, the second is 2, the third is 1, the fourth is 2, the fifth is 2, and the sixth is 1. The numbers are colored in a gradient from blue to red.

Ex-4

- Write a complete function that *accepts* two integers. It calculates 10% of first integer value, 20% of second integer value and *returns* float sum of two percentages to the calling statement in main().
- If $a = 100$, $b = 150$ were passed to the function, it returns the answer given below:

```
Final Answer = 40
```

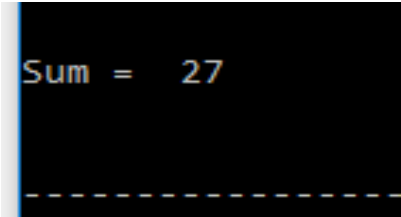
Ex-5

- An array (`arr`) has been initialized with the following values:

1	6	5	8	7	-1	8	5	10	7
---	---	---	---	---	----	---	---	----	---

Write a loop that calculates the total of the array values and displays the sum. The loop must stop when a -1 is read from the array

Output:

A screenshot of a terminal window with a black background. The text "Sum = 27" is displayed in a light blue or cyan monospaced font. Below the text, there is a dashed line, also in the same color, suggesting a prompt or a separator.

```
Sum = 27
```

Ex-6

```
void show_me(int n, int m)
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < m; j++)
            cout << "Q";
        cout << endl;
    }
}
```

Show the output of the following calls separately:

- i) show_me(7,3);
- ii) show_me(3,7);

Ex-7

- Your program has a global integer variable, `my_num` initialized to 120. A function (`f1`) adds 10 to the value of the global variable and return the answer every time it is called. Find the output of the following `main()` function:

```
int main()
{
    my_num = my_num+50;
    cout<<my_num<<endl<<endl;
    int ans;
    ans=f1();
    cout<<my_num<<endl<<endl;
    return 0;
}
```


Ex-8

- Write C++ code that asks the user to enter an integer. Let's assume that user enters a positive value. The code then uses a Boolean variable to decide if the entered value is even number or odd number.

```
Enter an integer : 31
Boolean Status = 0

-----
Process exited after 5.373
```

Solutions

Ex-1

```
int i = 0;
int num[3][3];
while(i < 3)
{
    int j = 0;
    while(j < 3)
    {
        cout << "i = " << i << " and j = " << j << " : ";
        cin >> num[i][j];
        j++;
    }
    i++;
}
```

Ex-2

```
cout<<endl<<"The Matrix is displyed below"<<endl;
for (int i = 0; i<3; i++)
{
    for (int j = 0; j<3; j++)
    {
        cout <<num[i][j]<<"\t";
    }
    cout<<endl;
}
...
```

Ex-3

```
int random[5]
for ( int i = 0; i < 5; i++ )
{
    random[i] = 1 + srand () % 2;
    cout<<random[i]<<endl;
}
```

Ex-4

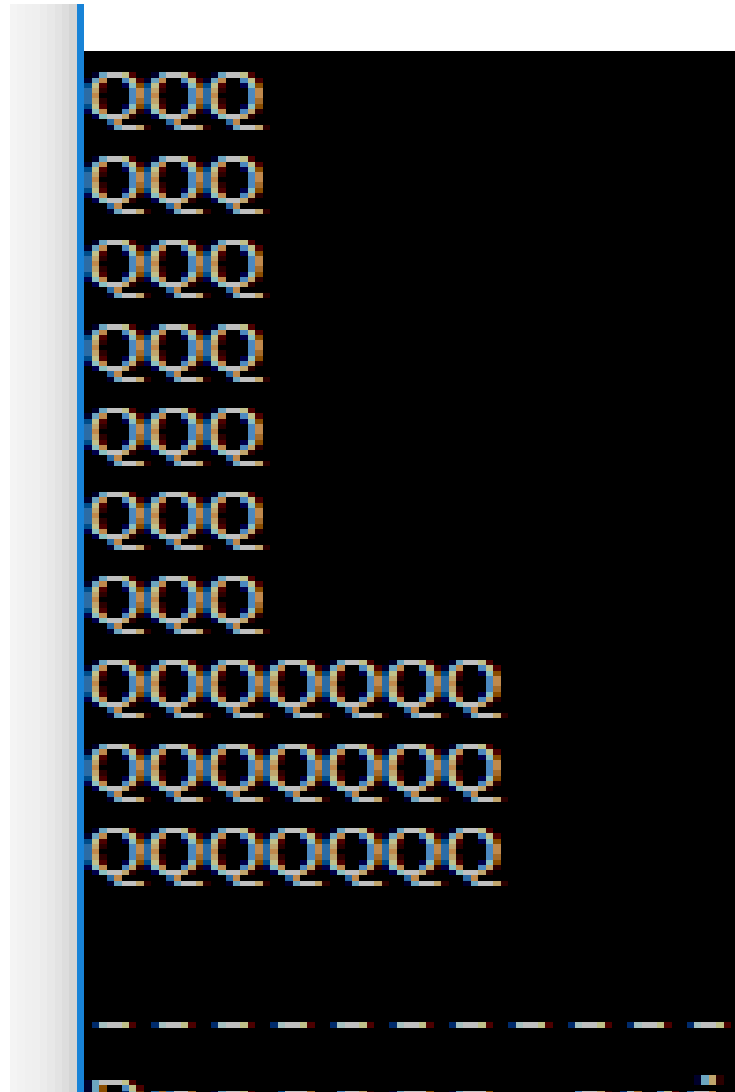
```
float my_func(int a, int b)
{
    float ans1,ans2;
    ans1 = a * 0.10;
    ans2 = b * 0.20;
    return ans1 + ans2;
}
```

Ex-5

```
int arr[10]= {1,6,5,8,7,-1,8,5,10,7};
int sum = 0;
cout<<endl<<endl;

for (int i = 0; i <10; i++)
{
    if (arr[i]>0)
    {
        sum = sum + arr[i];
    }
    else
    {
        break;
    }
}
cout<<"Sum = "<<sum<<endl<<endl;
```

Ex-6



Ex-7

170

180

Ex-8

```
int n;  
cout<<"Enter an integer : ";  
cin>>n;  
bool status;  
if (n % 2 == 0)  
    status = true; // The number is even if there  
else  
    status = false; // Otherwise, the number is o  
cout<<" Boolean Status = "<<status<<endl;
```